

Dual Slope ADC

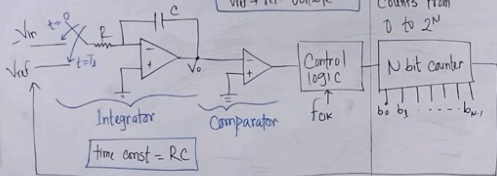
Dual Slope converter:

Adv: It requires small amount of circuit components. We can use many levels for quantization.

Disadvantage: Relatively slow.

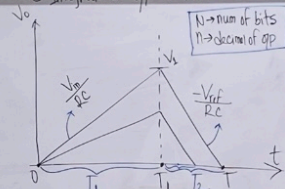
Dual Slope ADC

$V_{in} \rightarrow$ Analog i/p Voltage
 $V_{ref} \rightarrow$ ref. Voltage



i/p voltage range = $(V_{ref} - 0)$
 num of steps $\rightarrow 2^N$
 Resolution/Step size = $\frac{(V_{ref} - 0)}{2^N}$

⊗ Integrator's o/p $\rightarrow V_o$



$N \rightarrow$ num of bits
 $n \rightarrow$ decimal of o/p

here. $T_1 = 2^N T$
 $T_2 = n T$

$T \rightarrow$ Clock cycle/period
 $T = \frac{1}{f_{clk}}$

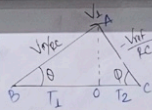
$0 < t < T_1 \rightarrow$ Integration phase = T_1
 $V_o = -\frac{V_{in}}{RC} t \rightarrow$ at $t = T_1 \rightarrow V_1 = -\frac{V_{in}}{RC} T_1$

Slope = $\frac{V_{in}}{RC}$

$T_1 < t < T \rightarrow$ Deintegration phase. $T_2 = T - T_1$
 $V_o = -\frac{V_{ref}}{RC} t \rightarrow$ Slope = $-\frac{V_{ref}}{RC}$

Total Reading time/Conversion time = $T_1 + T_2$

$\rightarrow$ Sampling Rate = $\frac{1}{T_1 + T_2}$
 $\rightarrow$ For Max Reading time $\Rightarrow T_1 = T_2 = 2^N T$
 $\therefore$ Max Reading time = $T_1 + T_2 = 2 \cdot 2^N T$
 $\rightarrow$ Max Sampling Rate = $\frac{1}{2 \cdot 2^N T}$



$\tan \theta = \frac{OA}{OB} = \frac{V_{in}}{RC} = \frac{V_1}{T_1} \quad \text{--- (i)}$

$\tan \phi = \frac{OA}{OC} = \frac{V_{ref}}{RC} = \frac{V_1}{T_2} \quad \text{--- (ii)}$

(i) - (ii) $\Rightarrow \frac{V_{in}}{V_{ref}} = \frac{T_2}{T_1} = \frac{n T}{2^N T}$

$\Rightarrow \frac{V_{in}}{V_{ref}} = \frac{n}{2^N}$

i/p $\rightarrow V_{in} = \frac{n}{2^N} V_{ref}$
 o/p $\rightarrow n = \frac{V_{in}}{V_{ref}} 2^N$

$V_{in} = \frac{n}{T_1 f_{clk}} V_{ref}$

Dual slope ADC → All formulas (Summarized)

Integration Phase (0 - T₁)

integrators opp → V_o(t)

$$V_o(t) = \frac{V_{in}}{RC} t \quad \left[\text{Slope} = \frac{V_{in}}{RC} \right]$$

$$\text{at } t = T_1 \rightarrow \left[V_o = \frac{V_{in}}{RC} T_1 \right]$$

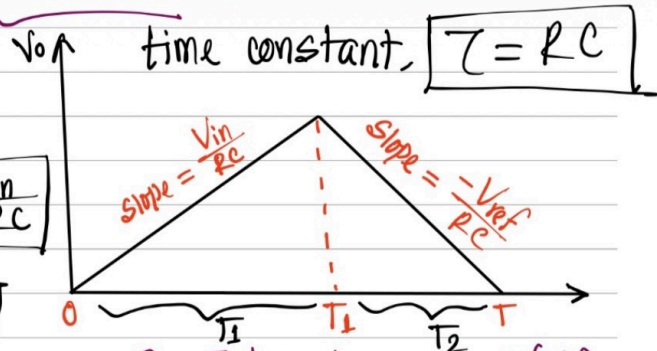


Fig: Integrators output (V_o)

De-integration Phase (T₁ - T)

$$V_o(t) = -\frac{V_{ref}}{RC} t \quad \left[\text{Slope} = -\frac{V_{ref}}{RC} \right]$$

T₁ → integration Phase Duration

T₂ → De-integration - - -

f_{clk} → Clock frequency (Hz)

$$\text{clock period, } T = \frac{1}{f_{clk}}$$

N → num of bits

n → Decimal of the opp.

$$\left[T_1 = 2^N T \right], \left[T_2 = n T \right], \text{Reading time/Conversion time, } T = T_1 + T_2$$

$$\rightarrow \left[\frac{V_{in}}{V_{ref}} = \frac{T_2}{T_1} \right] \quad \text{i/p} \rightarrow \left[V_{in} = \frac{n}{2^N} V_{ref} \right] \rightarrow \left[V_{in} = \frac{n}{T_1 f_{clk}} V_{ref} \right]$$

$$\text{o/p} \rightarrow \left[n = V_{in} \frac{2^N}{V_{ref}} \right] \rightarrow \text{decimal then convert into Binary}$$

→ i/p voltage Range = (V_{ref} - 0)

→ num of steps = 2^N

→ Resolution / step Size = $\frac{V_{ref} - 0}{2^N}$

→ Reading time = T₁ + T₂, Sampling Rate = $\frac{1}{T_1 + T_2}$

for Max Reading time → T₁ = T₂ = 2^N T ∴ max Reading time = T₁ + T₂ = 2 × 2^N T
 ∴ max Sampling Rate / Frequency = $\frac{1}{2 \times 2^N T}$

Q. A dual slope ADC has the reference voltage, $V_{ref} = 16\text{V}$. If the counter's output is 10110101 (in binary) and the clock period is 1.5MHz , what is the corresponding input voltage and conversion time.

Solⁿ: num of bit $N = 8$, $V_{ref} = 16\text{V}$
o/p, $n = \text{decimal of } (10110101)_2$
 $= 181$

$$f_{clk} = 1.5 \times 10^6, T = \frac{1}{f_{clk}}$$

$$\therefore \text{input Voltage, } V_{in} = \frac{n}{2^N} \times V_{ref}$$
$$= \frac{181}{2^8} \times 16$$

$$\boxed{V_{in} = 11.3125\text{V}}$$

$$T_1 = 2^N T = \frac{2^N}{f_{clk}} = \frac{2^8}{1.5 \times 10^6} = 0.171\text{ms}$$

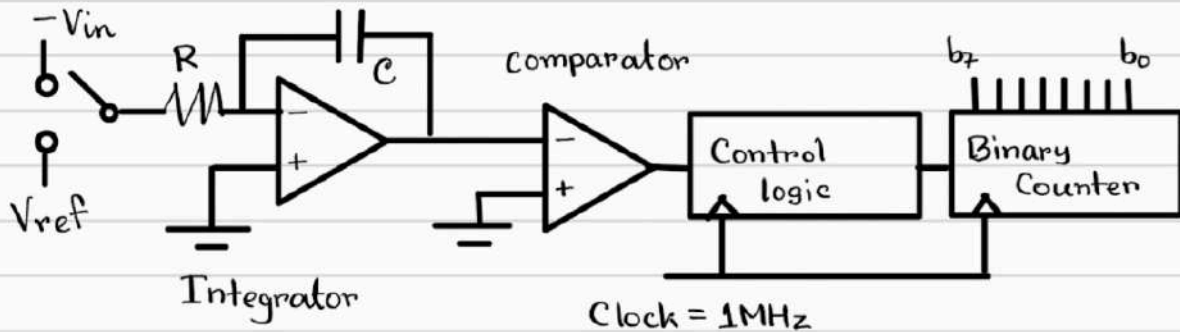
$$T_2 = nT = \frac{n}{f_{clk}} = \frac{181}{1.5 \times 10^6} = 0.121\text{ms}$$

$$\therefore \text{Total reading time / Conversion time} = T_1 + T_2$$
$$= 0.292\text{ms}$$

(Ans.)

Dual-Slope A/D Converter

- A dual slope A/D converter has $V_{ref} = 5V$, and a 8 bit counter that outputs an 8-bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits



- ① Determine the input voltage range, number of steps and step size (resolution) of the A/D converter.
- ② For a specific input voltage, the ADC outputs a count of $m=100$. Determine the input voltage.
- ③ Determine the total time required to get the reading in ②
- ④ Calculate the maximum sampling rate of the ADC.
- ⑤ A person wishes to get a sampling rate 4 times that of the current circuit. Determine how many bits should be used for the counter without changing any other system parameters.

(i) Input voltage range : $5V$ ($0 - V_{ref}$)
 number of steps $= 2^n = 2^8 = 256$
 step size $= \frac{\text{voltage range}}{\text{number of steps}} = 0.0195 V$

(ii) We know, $V_{in} = \frac{m}{2^n} V_{ref} = \frac{100}{256} \times 5V = 1.95V$

(iii) For a single conversion, the integrator

(a) integrates $-V_{in}$ for t_1 duration as counter counts from $0 - 2^n$

$$t_1 = \frac{2^n}{\text{clock rate}} = \frac{256}{1 \times 10^6} = 2.56 \times 10^{-4} s$$

(b) integrates V_{ref} for t_2 duration as counter counts from $0 - m$

$$t_2 = \frac{m}{\text{clock rate}} = \frac{100}{1 \times 10^6} = 1 \times 10^{-4} s$$

$$\text{total conversion time} = t_1 + t_2 = 3.56 \times 10^{-4} s$$

(iv) Maximum sampling rate is limited by the time required for worst case conversion of input $= 5V$

$$\text{In that case, } t_1 = t_2 = 2.56 \times 10^{-4} s$$

$$\text{thus, worst case conversion duration} = t_1 + t_2 = 5.12 \times 10^{-4} s$$

$$\text{Max sampling rate} = \frac{1}{5.12 \times 10^{-4} s} = 1953.125 \text{ Hz}$$

(v) Sampling duration $\propto 2^n$ // Sampling rate $\propto \frac{1}{2^n}$

$$\text{For 4 times sampling rate, counter bit} = (8 - 2) = 6$$

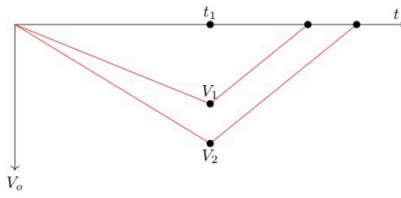


Figure 1.15: Integrator output vs time

For a 6-bit Dual Slope ADC $V_1 = -6$ V, $V_2 = -8$ V in figure 1.15 and the maximum allowable sampling frequency is 1.5625 kHz. The time constant τ of the integrator is 0.5 ms.

(a)	Find the two input voltages.
(b)	If the reference voltage is 15 V, find the digital outputs for the inputs found in (a).
(c)	Find the encoded output for which the slope of the integration phase of the integrator would be the average of the two given.

Solution:

(a)

$$V_{in_1} = V_1 \times \frac{RC}{t_1} = -(-6) \times \frac{0.5 \times 10^{-3}}{t_1}$$

Now, worst case conversion time, $2t_1 = \frac{1}{1.5625 \times 10^3} = 0.64$ ms, $\therefore t_1 = 0.32$ ms.

$$\therefore V_{in_1} = \frac{3 \times 10^{-3}}{0.32 \times 10^{-3}} = 9.375$$

Similarly,

$$V_{in_2} = \frac{-(-8) \times 0.5 \times 10^{-3}}{0.32 \times 10^{-3}} = 12.5$$

(b)

$$N_1 + 1 = \frac{V_{in_1}}{V_{ref}} \times 2^6 = \frac{9.375}{15} \times 64 = 40 \Rightarrow N_1 = 39$$

$$N_2 + 1 = \frac{V_{in_2}}{V_{ref}} \times 2^6 = \frac{12.5}{15} \times 64 = 53(\text{floored value}) \Rightarrow N_2 = 52$$

$$\therefore D_1 = \text{binary of } N_1 = 100111$$

$$\therefore D_2 = \text{binary of } N_2 = 110100$$

(c) Let slopes of integration for V_{in_1} & V_{in_2} are m_1 & m_2 respectively.

$$m_1 = -\frac{V_{in_1}}{RC} = -\frac{9.375}{0.5} = -18.75 \text{ V/ms}$$

$$m_2 = -\frac{V_{in_2}}{RC} = -\frac{12.5}{0.5} = -25 \text{ V/ms}$$

$$\therefore m = \frac{m_1 + m_2}{2} = -21.875 \text{ V/ms} \Rightarrow -\frac{V_{in}}{RC} = -21.875 \Rightarrow V_{in} = 0.5 \times 21.875 = 10.9375$$

$$\therefore N + 1 = \frac{V_{in}}{V_{ref}} \times 2^6 = \frac{10.9375}{15} \times 64 = 46(\text{floored value}) \Rightarrow N = 45$$

$$\therefore D = \text{binary of } 45 = 101101$$